

Module-5

Uncertain Knowledge and Reasoning

**Quantifying Uncertainty- Bayes Rule -
Bayesian Belief Network- Approximate
Inference in Bayesian networks**

QUANTIFYING UNCERTAINTY

Outline

- **Introduction**
- **Uncertainty**
- **Probability**
- **Syntax and Semantics**
- **Inference**
- **Conditional Independence**
- **Bayes' Rule**

Uncertain knowledge and reasoning

- Uncertainty: Till now, we have learned knowledge representation using first-order logic and propositional logic with certainty, which means we were sure about the predicates. ... **So to represent uncertain knowledge, where we are not sure about the predicates, we need uncertain reasoning or probabilistic reasoning.**

KB systems. Medical example.

We want to build a KB system for the **diagnosis of pneumonia**.

Problem description:

- **Disease:** pneumonia
- **Patient symptoms (findings, lab tests):**
 - Fever, Cough, Paleness, WBC (white blood cells) count, Chest pain, etc.

Representation of a patient case:

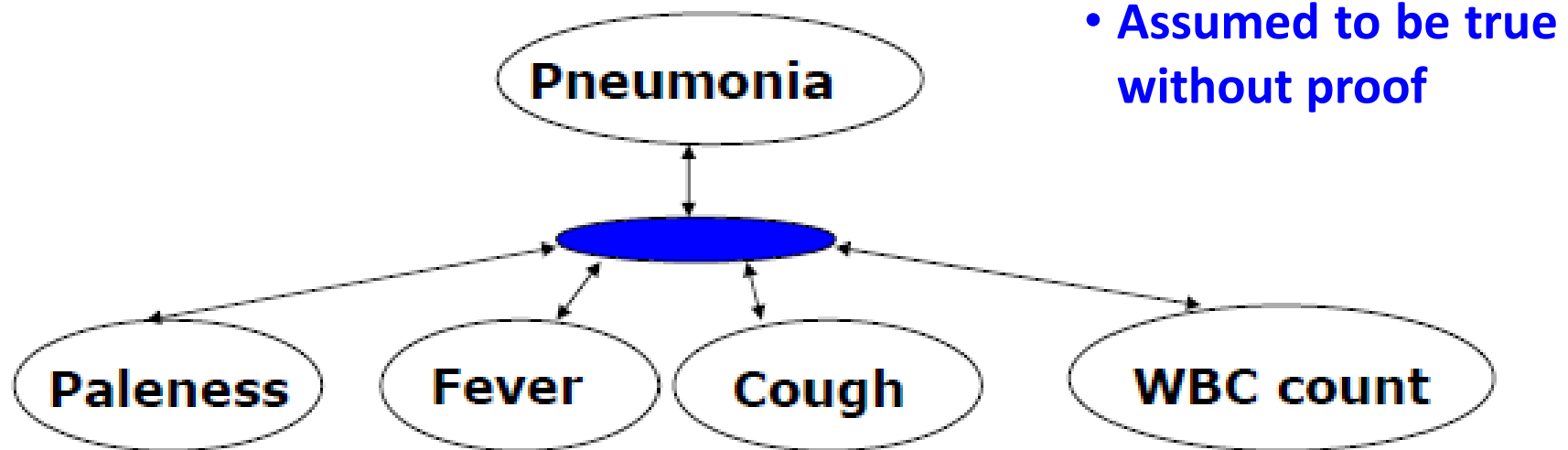
- Statements that hold (are true) for the patient.

E.g: Fever = *True*
 Cough = *False*
 WBCcount = *High*

Diagnostic task: we want to decide whether the patient suffers from the pneumonia or not given the symptoms

Uncertainty

To make diagnostic inference possible we need to represent knowledge (axioms) that relate symptoms and diagnosis



Problem: disease/symptoms relations are not deterministic

- They are **uncertain (or stochastic)** and vary from patient to patient

The stochastic environment is random in nature which is not unique and cannot be completely determined by the agent.

- **P4** medicine refers to a proactive approach to healthcare focusing on Personalized, Predictive, Preventive, and Participatory medicine.

P4 Medicine

● PREDICT ● PREVENT ● PERSONALIZE ● PARTICIPATE



Uncertainty

Why are relations uncertain?

- **Observability**
 - It is impossible to observe all relevant components of the world
 - Observable components behave stochastically even if the underlying world is deterministic
- **Efficiency, capacity limits**
 - It is often impossible to enumerate and model all components of the world and their relations
 - abstractions can become stochastic

Humans can reason with uncertainty !!!

- Can computer systems do the same?

Methods for representing uncertainty

Extensions of the propositional and first-order logic

- Use, uncertain, imprecise statements (relations)

Example: Propositional logic with certainty factors

Very popular in 70-80s in knowledge-based systems (MYCIN)

- **Facts (propositional statements)** are assigned a **certainty value** reflecting the belief in that the statement is satisfied:

$$CF(Pneumonia = True) = 0.7$$

- **Knowledge:** typically in terms of **modular rules**

If	1. The patient has cough, and 2. The patient has a high WBC count, and 3. The patient has fever
Then	with certainty 0.7 the patient has pneumonia

CF-Cystic fibrosis

Methods for representing uncertainty

Probability theory

- A well defined theory for modeling and reasoning in the presence of uncertainty
- A natural choice to replace certainty factors

Facts (propositional statements)

- Are represented via **random variables** with two or more values

Example: *Pneumonia* is a random variable

values: *True* and *False*

- Each value can be achieved **with some probability:**

$$P(Pneumonia = True) = 0.001$$

$$P(WBCcount = high) = 0.005$$

- Introduction to reasoning under uncertainty
- Review of probability
 - Axioms and inference
 - Conditional probability
 - Probability distributions

Uncertainty

- Back to planning:

- Let action $A(t)$ denote leaving for the airport t minutes before the flight
- For a given value of t , will $A(t)$ get me there on time?

- Problems:

- Partial observability (roads, other drivers' plans, etc.)
- Noisy sensors (traffic reports)
- Uncertainty in action outcomes (flat tire, etc.)
- Immense complexity of modeling and predicting traffic

How To Deal With Uncertainty

- Implicit methods:

- Ignore uncertainty as much as possible
- Build procedures that are robust to uncertainty
- This is the approach in the planning methods studied so far (e.g. monitoring and replanning)

- *Explicit methods*

- Build a *model* of the world that *describes the uncertainty* (about the system's state, dynamics, sensors, model)
- Reason about the effect of actions given the model

How To Represent Uncertainty

- What language should we use?
- What are the semantics of our representations?
- What queries can we answer with our representations?
- How do we answer them?
- How do we construct a representation?
- Do we need to ask an expert or can we learn from data?

Handling Uncertain Knowledge

Why Not Use First-Order Logic?

Problems using first-order logic for diagnosis:

Laziness:

Too much work to make complete rules. Too much work to use them

Theoretical ignorance:

Complete theories are rare

Practical ignorance:

We can't run all tests anyway

Probability can be used to *summarize* the laziness and ignorance !

Why Not Use First-Order Logic?

- A purely logical approach has two main problems:
 - Risks falsehood
 - * $A(25)$ will get me there on time.
 - Leads to conclusions that are too weak:
 - * $A(25)$ will get me there on time if there is no accident on the bridge and it does not rain and my tires remain intact, etc. etc.
 - * $A(1440)$ might reasonably be said to get me there on time (but I would have to stay overnight at the airport!)

Using FOL for (Medical) Diagnosis

$\forall p \text{ Symptom}(p, \text{Toothache}) \Rightarrow \text{Disease}(p, \text{Cavity})$

Not correct...

$\forall p \text{ Symptom}(p, \text{Toothache}) \Rightarrow \text{Disease}(p, \text{Cavity})$
 $\vee \text{Disease}(p, \text{GumDisease}) \vee \text{Disease}(p, \text{WisdomTooth}) \vee \text{Disease}(p, \text{Abscess})$

Not complete...

$\forall p \text{ Disease}(p, \text{Cavity}) \Rightarrow \text{Symptom}(p, \text{Toothache})$

Not correct...

Methods for Handling Uncertainty

- *Default (non-monotonic) logic*: make assumptions unless contradicted by evidence.
 - E.g. “Assume my car doesn't have a flat tire.”
 - What assumptions are reasonable? What about contradictions?
- *Rules with fudge factor*:
 - E.g. “Sprinkler $\rightarrow_{0.99}$ WetGrass”, “WetGrass $\rightarrow_{0.7}$ Rain”
 - But: Problems with combination (e.g. Sprinkler causes rain?)
- *Probability*:
 - E.g. Given what I know, A(25) succeed with probability 0.2
- *Fuzzy logic*:
 - E.g. WetGrass is true to degree 0.2
 - But: Handles degree of truth, NOT uncertainty.

Probability

- A well-known and well-understood framework for dealing with uncertainty
- Has a clear semantics
- Provides principled answers for:
 - Combining evidence
 - Predictive and diagnostic reasoning
 - Incorporation of new evidence
- Can be learned from data
- Intuitive to human experts (arguably?)
 - Model agent's degree of belief
 - Given the available evidence,
 - A_{25} will get me there on time with probability 0.04

Beliefs (Bayesian Probabilities)

- We use probability to describe uncertainty due to:
 - Laziness: failure to enumerate exceptions, qualifications etc.
 - Ignorance: lack of relevant facts, initial conditions etc.
 - True randomness? Quantum effects? ...
- *Beliefs (Bayesian or subjective probabilities)* relate propositions to one's current state of knowledge
 - E.g. $P(A(25) | \text{no reported accident}) = 0.1$

These are *not assertions about the world / absolute truth*
- Beliefs change with new evidence:
 - E.g. $P(A(25) | \text{no reported accident, 5am}) = 0.2$
- This is analogous to logical entailment: $KB \models \alpha$ means that α is true *given* the KB , but may not be true in general.

Rational Decisions

A rational decision must consider:

- The relative importance of the sub-goals
- Utility theory
 - The degree of belief that the sub-goals will be achieved
- Probability theory

Decision theory = probability theory + utility theory :

*the agent is rational **if and only if** it chooses the action that yields the highest expected utility, averaged over all possible outcomes of the action”*

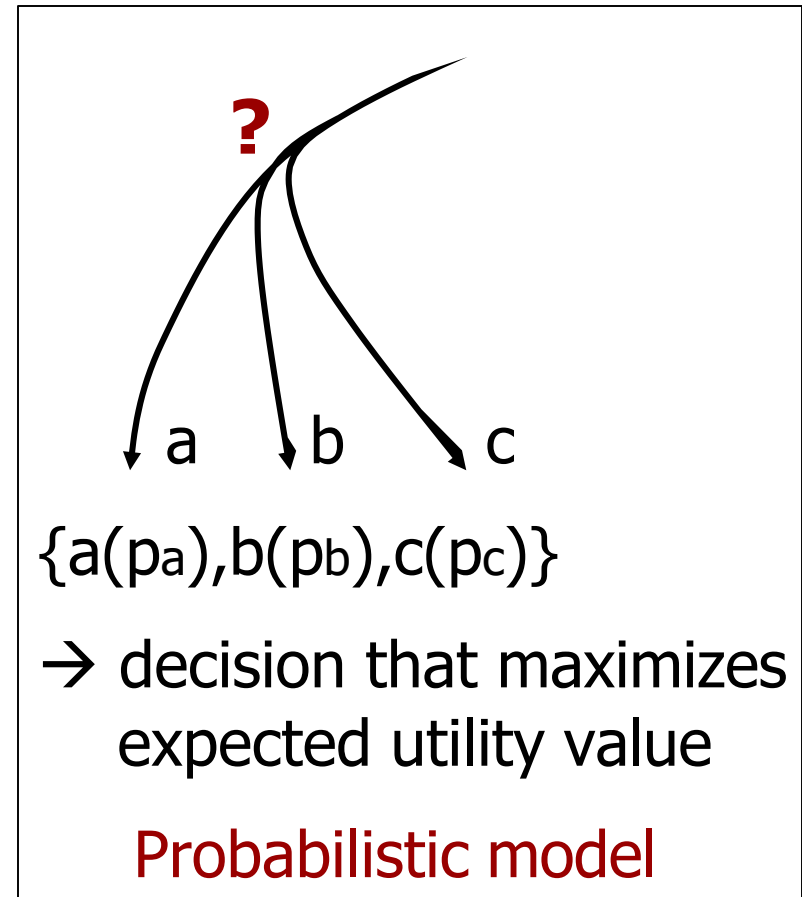
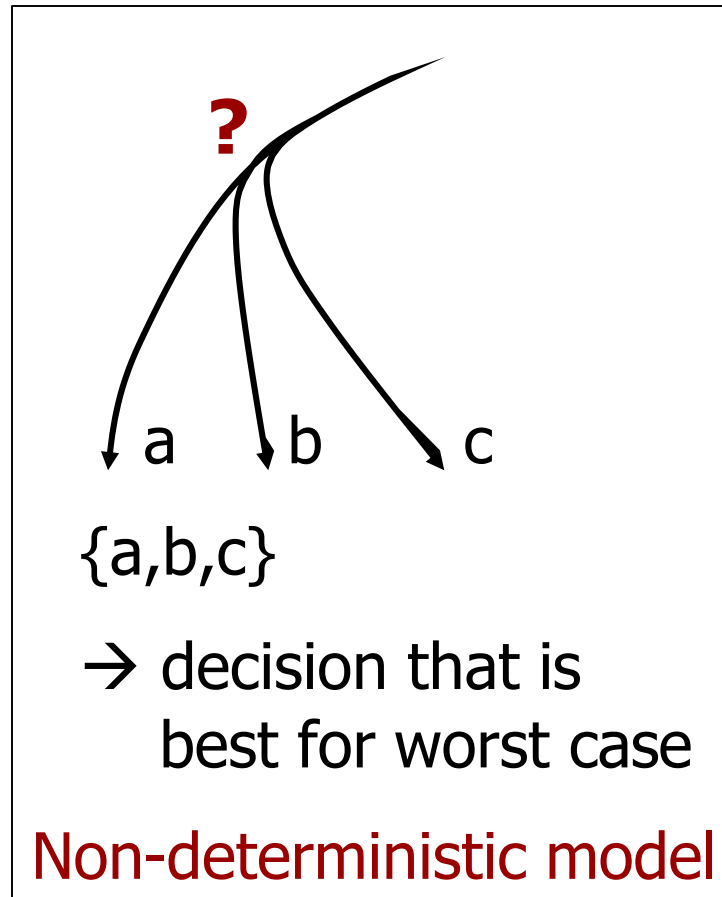
Combining Beliefs & Desires under Uncertainty

- Modern agents talk of utility rather than good and evil, but the principle is the same.
- Agent's preferences between world states are captured by a **Utility Function** $U(S)$
- It assigns a single number to express the desirability of a state.
- Utilities are combined with outcome probabilities of actions to give an expected utility for each action.
- A nondeterministic **action** A will have possible outcome states $Result_i(A)$ where the index i ranges over the different outcomes.
- Prior to the execution of A , the agent assigns probability $P(Result_i(A) | Do(A), E)$ to each outcome, where E summarizes the agent's available evidence about the world and $Do(A)$ is the proposition that action A is executed in the current state.
- The Expected Utility of A is

$$EU[A] = \sum_{i=1, \dots, n} p(Result_i(A) | Do(A)) U(Result_i)$$

- The principle of maximum expected utility (MEU) says that a rational agent should choose an action

Non-Deterministic vs. Probabilistic Uncertainty

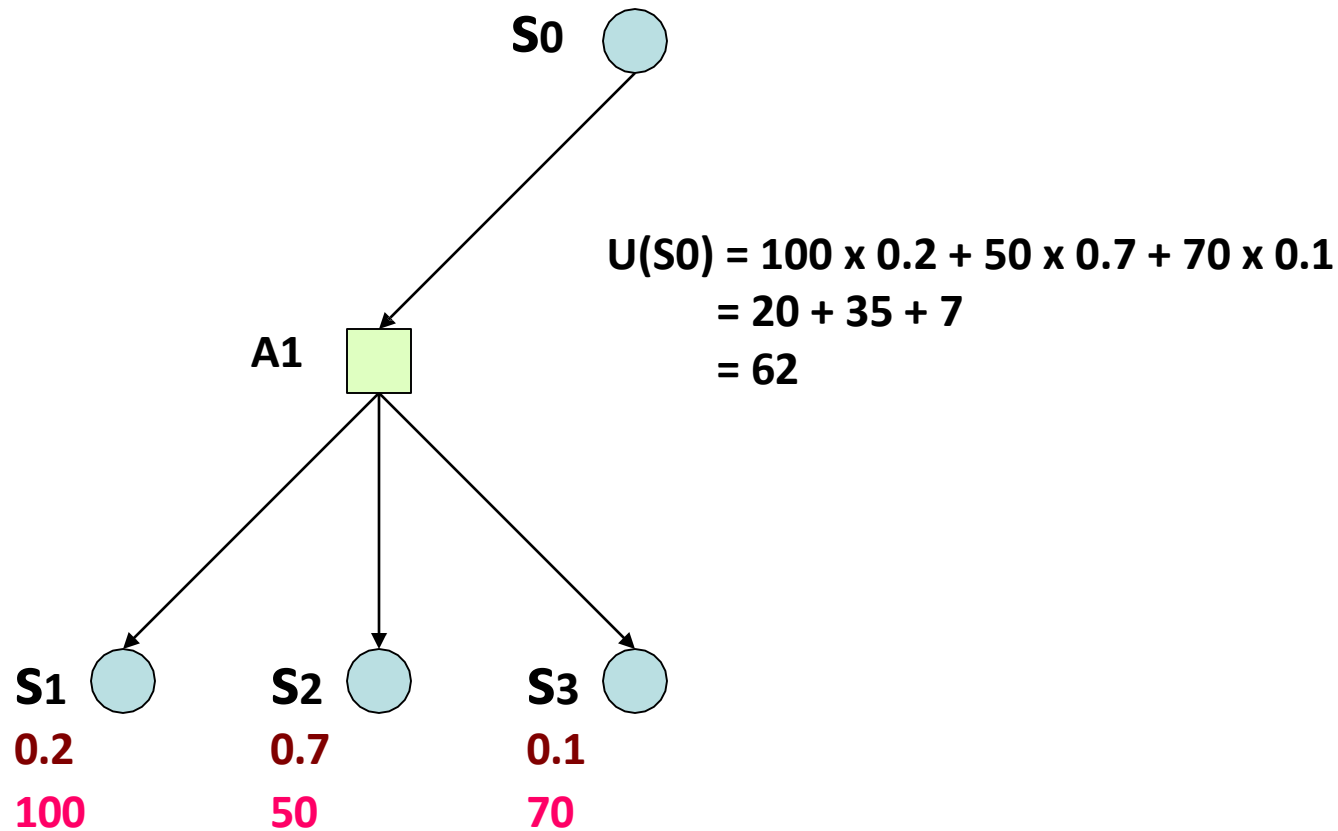


~ Adversarial search

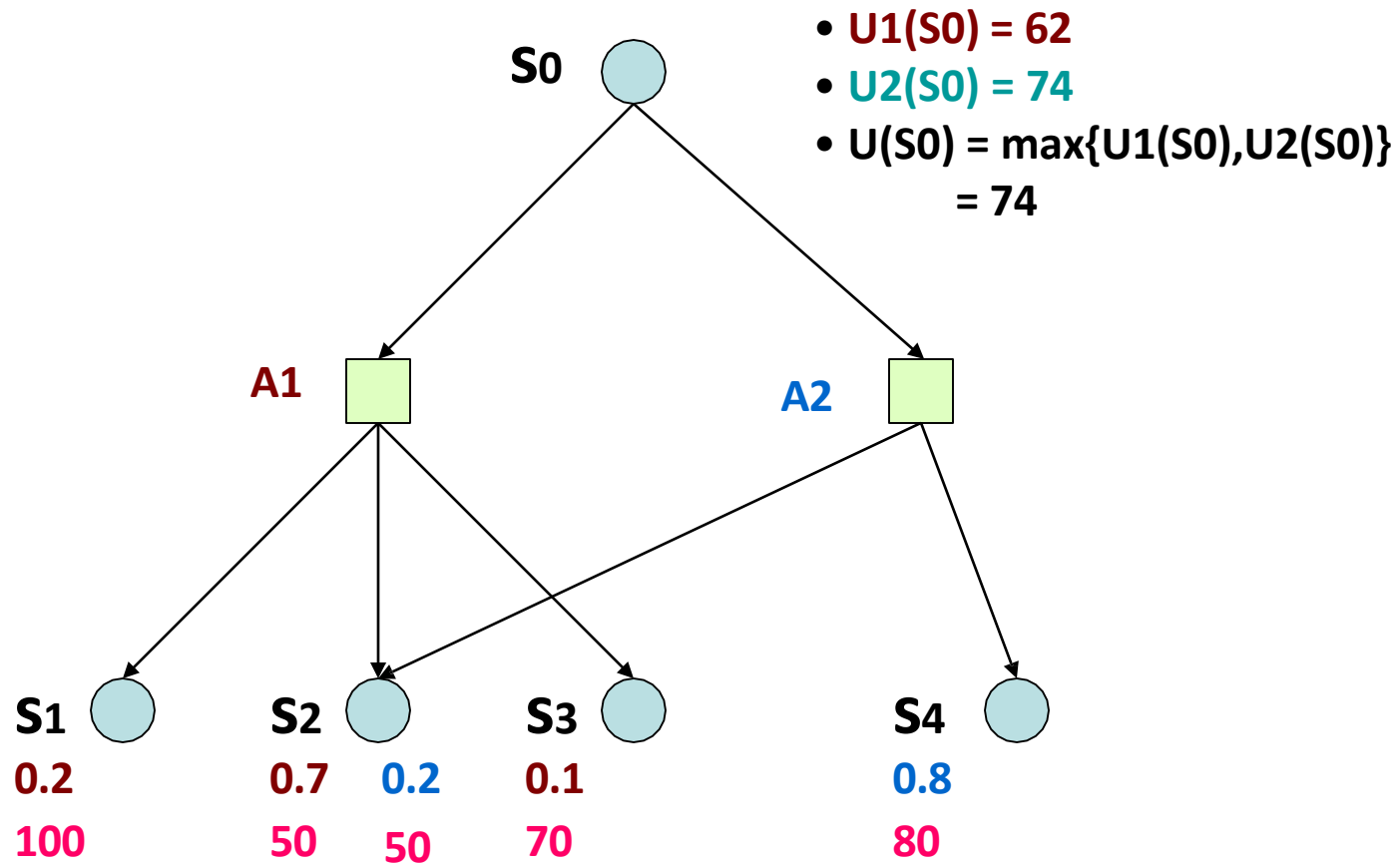
Expected Utility

- Random variable X with n values $x_1, \dots, x_n$ and distribution $(p_1, \dots, p_n)$
E.g.: X is the state reached after doing an action A under uncertainty
- Function U of X
E.g., U is the utility of a state
- The expected utility of A is
$$EU[A] = \sum_{i=1}^n p(x_i|A) U(x_i)$$

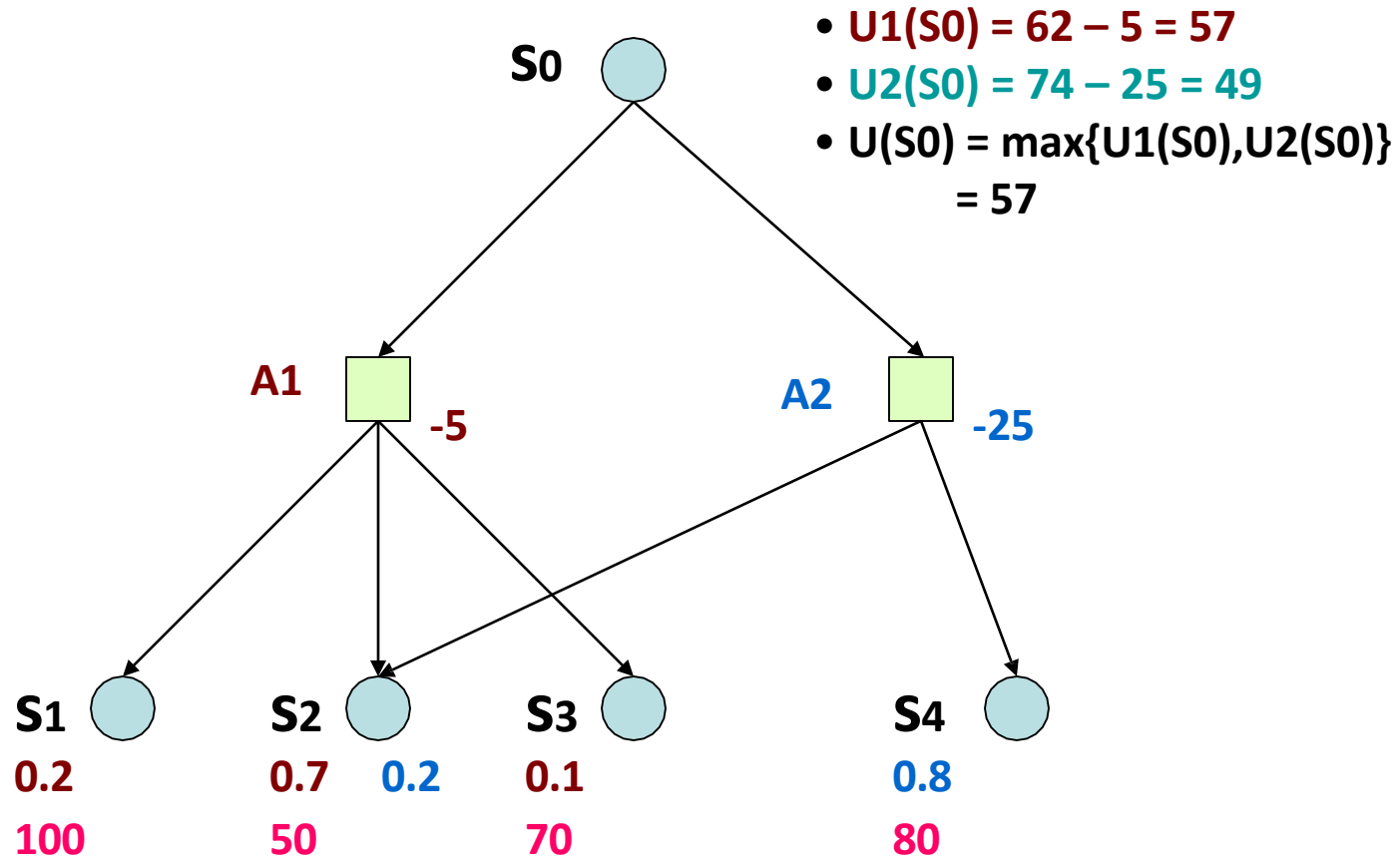
One State/One Action Example



One State/Two Actions Example



Introducing Action Costs



Maximum Expected Utility (MEU) Principle

- A **rational agent** should choose the action that maximizes agent's expected utility
- This is the basis of the field of **decision theory**
- The MEU principle provides a **normative criterion** for rational choice of action

Design for decision-theoretic agent

- **Function of a decision-theoretic agent that selects rational actions.**

function DT-AGENT(*percept*) **returns** an *action*

persistent: *belief state*, probabilistic beliefs about the current state of the world
action, the agent's action

update *belief state* based on *action* and *percept*

calculate outcome probabilities for actions,

given action descriptions and current *belief state*

select *action* with highest expected utility

given probabilities of outcomes and utility information

return *action*

Probability

A measure of how likely it is that some event will occur; a number expressing the ratio of favorable cases to the whole number of cases possible

Probabilistic assertions **summarize** effects of

- **laziness**: failure to enumerate exceptions, qualifications, etc.
- **Theoretical and practical ignorance**: lack of relevant facts, initial conditions, etc.

Subjective probability:

- Probabilities relate propositions to agent's own state of knowledge

$$\text{e.g., } P(A_{25} \mid \text{no reported accidents}) = 0.06$$

These are **not** assertions about the world

Probabilities of propositions change with new evidence:

$$\text{e.g., } P(A_{25} \mid \text{no reported accidents, 5 a.m.}) = 0.15$$

Probability

Probabilities are either:

- **Prior probability** (**unconditional**)
 - Before any evidence is obtained
- **Posterior probability** (**conditional**)
 - After evidence is obtained

Making Decisions Under Uncertainty

- Suppose I believe the following:
 $P(A(25) \text{ gets me there on time} \mid \dots) = 0.04$
 $P(A(90) \text{ gets me there on time} \mid \dots) = 0.70$
 $P(A(120) \text{ gets me there on time} \mid \dots) = 0.95$
 $P(A(1440) \text{ gets me there on time} \mid \dots) = 0.9999$
- Which action should I choose?
 - Depends on my preferences for missing flight vs. airport cuisine, etc.
 - *Utility theory* is used to represent and infer preferences.
 - *Decision theory* = utility theory + probability theory

Random Variables

- A *random variable* X describes an outcome that cannot be determined in advance
 - E.g. The roll of a die
 - E.g. Number of e-mails received in a day
- The *sample space (domain)* S of a random variable X is the set of all possible values of the variable
 - E.g. For a die, $S = \{1, 2, 3, 4, 5, 6\}$
 - E.g. For number of emails received in a day, S is the natural numbers
- An *event* is a subset of S .
 - E.g. $e = \{1\}$ corresponds to a die roll of 1
 - E.g. number of e-mails in a day more than 100

Syntax

- Basic element: **random variable**
- Similar to propositional logic: possible worlds defined by assignment of values to random variables.
- **Boolean** random variables
e.g., *Cavity* (do I have a cavity?)
- **Discrete** random variables –finite number of value
e.g., *Weather* is one of $\langle \text{sunny}, \text{rainy}, \text{cloudy}, \text{snow} \rangle$
- Domain values must be exhaustive and mutually exclusive
- Elementary proposition constructed by assignment of a value to a random variable:
e.g., *Weather* = *sunny*, *Cavity* = *false* (abbreviated as $\neg \text{cavity}$)
- Complex propositions formed from elementary propositions and standard logical connectives
e.g., *Weather* = *sunny* $\vee$ *Cavity* = *false*

Syntax

- **Atomic event:** A complete specification of the state of the world about which the agent is uncertain

E.g., if the world consists of only two Boolean variables *Cavity* and *Toothache*, then there are 4 distinct atomic events:

$Cavity = false \wedge Toothache = false$

$Cavity = false \wedge Toothache = true$

$Cavity = true \wedge Toothache = false$

$Cavity = true \wedge Toothache = true$

- **Atomic events are mutually exclusive and exhaustive**

Probability for Discrete Random Variables

- Usually, random variables are governed by some “law of nature”, described as a *probability function* P defined on S .
- $P(x)$ defines the chance that variable X takes value $x \in S$.

E.g. for a die roll with a fair die, $P(1) = P(2) = \dots = P(6) = 1/6$

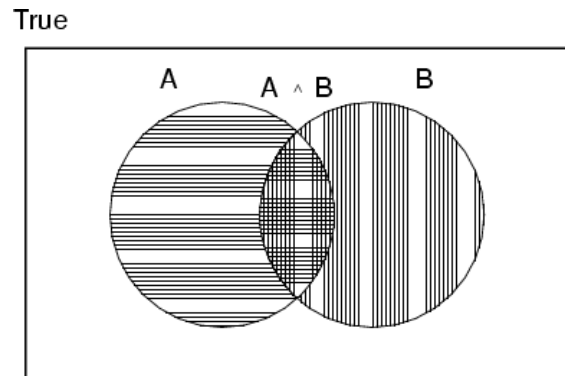
- Note that we still cannot determine the value of X , just the chance of encountering a given value
- If X is a discrete variable, then a probability space $P(x)$ has the following properties:

$$0 \leq P(x) \leq 1, \forall x \in S \text{ and } P(x) = 1, x \in S$$

$\in$ means “is an element of ()”

Axioms of probability

- For any propositions A, B
 - $0 \leq P(A) \leq 1$
 - $P(\text{true}) = 1$ and $P(\text{false}) = 0$
 - $P(A \vee B) = P(A) + P(B) - P(A \wedge B)$



- These three axioms are often called **Kolmogorov's axioms**.

Prior probability

- **Prior or unconditional probabilities** of propositions
e.g., $P(\text{Cavity} = \text{true}) = 0.144$ and $P(\text{Weather} = \text{sunny})$ correspond to belief prior to arrival of any (new) evidence
- **Probability distribution** gives values for all possible assignments:
 $\mathbf{P}(\text{Weather}) = \langle 0.72, 0.1, 0.08, 0.1 \rangle$ (**normalized**, i.e., sums to 1)
- **Joint probability distribution** for a set of random variables gives the **probability of every atomic event** on those random variables
- $\mathbf{P}(\text{Weather}, \text{Cavity})$ = a 4×2 matrix of values:

<i>Weather</i> =	sunny	rainy	cloudy	snow
<i>Cavity</i> = true	0.144	0.02	0.016	0.02
<i>Cavity</i> = false	0.576	0.08	0.064	0.08

- Every question about a domain can be answered by the joint distribution

Conditional probability

- **Conditional or posterior probabilities**

e.g., $P(\text{cavity} \mid \text{toothache}) = 0.8$

i.e., given that *toothache* is all I know

- (Notation for conditional distributions:

$\mathbf{P}(\text{Cavity} \mid \text{Toothache}) = 2\text{-element vector of } 2\text{-element vectors})$

- If we know more, e.g., *cavity* is also given, then we have

$P(\text{cavity} \mid \text{toothache}, \text{catch}) = 1$

- New evidence may be irrelevant, allowing simplification, e.g.,

$P(\text{cavity} \mid \text{toothache}, \text{sunny}) = P(\text{cavity} \mid \text{toothache}) = 0.8$

- This kind of inference, sanctioned by domain knowledge, is crucial

Conditional probability

- Definition of **conditional probability**:

$$P(a \mid b) = P(a \wedge b) / P(b) \text{ if } P(b) > 0$$

- Product rule** gives an alternative formulation:

$$P(a \wedge b) = P(a \mid b) P(b) = P(b \mid a) P(a)$$

- A general version holds for whole distributions, e.g.,

$$\mathbf{P}(\textit{Weather}, \textit{Cavity}) = \mathbf{P}(\textit{Weather} \mid \textit{Cavity}) \mathbf{P}(\textit{Cavity})$$

- (View as a set of 4×2 equations, **not** matrix mult.)

- Chain rule** is derived by successive application of product rule:

$$\begin{aligned} \mathbf{P}(X_1, \dots, X_n) &= \mathbf{P}(X_1, \dots, X_{n-1}) \mathbf{P}(X_n \mid X_1, \dots, X_{n-1}) \\ &= \mathbf{P}(X_1, \dots, X_{n-2}) \mathbf{P}(X_{n-1} \mid X_1, \dots, X_{n-2}) \mathbf{P}(X_n \mid X_1, \dots, X_{n-1}) \\ &= \dots \\ &= \pi_{i=1}^n \mathbf{P}(X_i \mid X_1, \dots, X_{i-1}) \end{aligned}$$

Or

$$P(A_1, A_2, \dots, A_n) = P(A_1 \mid A_2, \dots, A_n) P(A_2 \mid A_3, \dots, A_n) P(A_{n-1} \mid A_n) P(A_n)$$

Inference by Numeration

- Start with the **joint probability distribution**:

	<i>toothache</i>		$\neg$ <i>toothache</i>	
	<i>catch</i>	$\neg$ <i>catch</i>	<i>catch</i>	$\neg$ <i>catch</i>
<i>cavity</i>	.108	.012	.072	.008
$\neg$ <i>cavity</i>	.016	.064	.144	.576

- For any proposition ϕ , sum the atomic events where it is true:
$$P(\phi) = \sum_{\omega: \omega \models \phi} P(\omega)$$

Inference by enumeration

- Start with the joint probability distribution:

	<i>toothache</i>		$\neg$ <i>toothache</i>	
	<i>catch</i>	$\neg$ <i>catch</i>	<i>catch</i>	$\neg$ <i>catch</i>
<i>cavity</i>	.108	.012	.072	.008
$\neg$ <i>cavity</i>	.016	.064	.144	.576

- For any proposition ϕ , sum the atomic events where it is true:
 $P(\phi) = \sum_{\omega: \omega \models \phi} P(\omega)$
- $P(\text{toothache}) = 0.108 + 0.012 + 0.016 + 0.064 = 0.2$

Inference by enumeration

- Start with the joint probability distribution:

	<i>toothache</i>		$\neg$ <i>toothache</i>	
	<i>catch</i>	$\neg$ <i>catch</i>	<i>catch</i>	$\neg$ <i>catch</i>
<i>cavity</i>	.108	.012	.072	.008
$\neg$ <i>cavity</i>	.016	.064	.144	.576

- Can also compute conditional probabilities:

$$\begin{aligned}
 P(\neg \text{cavity} \mid \text{toothache}) &= \frac{P(\neg \text{cavity} \wedge \text{toothache})}{P(\text{toothache})} \\
 &= \frac{0.016 + 0.064}{0.108 + 0.012 + 0.016 + 0.064} \\
 &= 0.4
 \end{aligned}$$

$$P(a \mid b) = P(a \wedge b) / P(b) \text{ if } P(b) > 0$$

Inference by enumeration

	<i>toothache</i>		$\neg$ <i>toothache</i>	
	<i>catch</i>	$\neg$ <i>catch</i>	<i>catch</i>	$\neg$ <i>catch</i>
<i>cavity</i>	.108	.012	.072	.008
$\neg$ <i>cavity</i>	.016	.064	.144	.576

- Can also compute conditional probabilities:

$$\begin{aligned} P(\text{cavity} \mid \text{toothache}) &= \frac{P(\text{cavity} \wedge \text{toothache})}{P(\text{toothache})} \\ &= \frac{0.108 + 0.012}{0.108 + 0.012 + 0.016 + 0.064} \\ &= 0.6 \end{aligned}$$

Inference by enumeration

	<i>toothache</i>		$\neg$ <i>toothache</i>	
	<i>catch</i>	$\neg$ <i>catch</i>	<i>catch</i>	$\neg$ <i>catch</i>
<i>cavity</i>	.108	.012	.072	.008
$\neg$ <i>cavity</i>	.016	.064	.144	.576

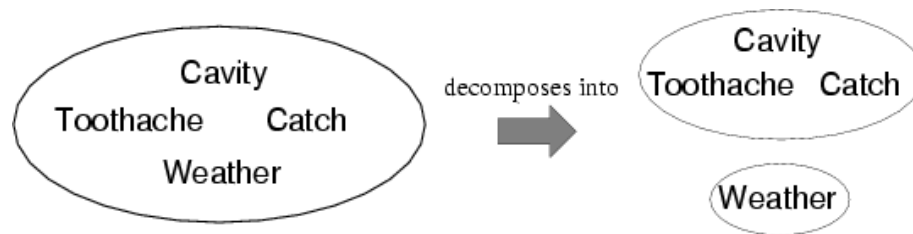
$$\begin{aligned} P(\text{Cavity} \vee \text{toothache}) \\ &= 0.108 + 0.012 + 0.072 + 0.008 + 0.016 + 0.064 \\ &= 0.28 \end{aligned}$$

0.108 + 0.012 – no need to add two times

Independence

- A and B are independent iff

$$\mathbf{P}(A|B) = \mathbf{P}(A) \quad \text{or} \quad \mathbf{P}(B|A) = \mathbf{P}(B) \quad \text{or} \quad \mathbf{P}(A, B) = \mathbf{P}(A) \mathbf{P}(B)$$



$$\begin{aligned} &\mathbf{P}(\textit{Toothache}, \textit{Catch}, \textit{Cavity}, \textit{Weather}) \\ &= \mathbf{P}(\textit{Toothache}, \textit{Catch}, \textit{Cavity}) \mathbf{P}(\textit{Weather}) \end{aligned}$$

- 32 entries($4 \times 2 \times 2 \times 2$) reduced to 8; for n independent biased coins,
- $O(2^n) \rightarrow O(n)$
- Absolute independence powerful but rare
- Dentistry is a large field with hundreds of variables, none of which are independent. What to do?
- $\mathbf{P}(\textit{Weather}, \textit{Cavity}) =$ a 4×2 matrix of values:

<i>Weather</i> =	sunny	rainy	cloudy	snow
<i>Cavity</i> = true	0.144	0.02	0.016	0.02
<i>Cavity</i> = false	0.576	0.08	0.064	0.08

Conditional independence

- $\mathbf{P}(\textit{Toothache}, \textit{Cavity}, \textit{Catch})$ has $2^3 - 1 = 7$ independent entries
- If I have a cavity, the probability that the probe catches in it doesn't depend on whether I have a toothache:
(1) $\mathbf{P}(\textit{catch} \mid \textit{toothache}, \textit{cavity}) = \mathbf{P}(\textit{catch} \mid \textit{cavity})$
- The same independence holds if I haven't got a cavity:
(2) $\mathbf{P}(\textit{catch} \mid \textit{toothache}, \neg \textit{cavity}) = \mathbf{P}(\textit{catch} \mid \neg \textit{cavity})$
- *Catch* is **conditionally independent** of *Toothache* given **Cavity**:
 $\mathbf{P}(\textit{Catch} \mid \textit{Toothache}, \textit{Cavity}) = \mathbf{P}(\textit{Catch} \mid \textit{Cavity})$
- Equivalent statements:
 $\mathbf{P}(\textit{Toothache} \mid \textit{Catch}, \textit{Cavity}) = \mathbf{P}(\textit{Toothache} \mid \textit{Cavity})$
 $\mathbf{P}(\textit{Toothache}, \textit{Catch} \mid \textit{Cavity}) = \mathbf{P}(\textit{Toothache} \mid \textit{Cavity}) \mathbf{P}(\textit{Catch} \mid \textit{Cavity})$
- Conditional independence states that two events, A and B, are independent of each other when a third event, C, is known or given to have occurred.
- **Conditional Probability**: The likelihood of an event occurring given that **another event has already occurred**.
- **Independence**: The occurrence of **one event does not affect the probability of another**.

Bayes' Rule

- Product rule $P(A \wedge B) = P(A \mid B) P(B) = P(B \mid A) P(A)$

$\Rightarrow$ Bayes' rule:

$$P(A \mid B) = \frac{P(B \mid A) \cdot P(A)}{P(B)}$$

Bayes' rule determines the probability of an event when knowledge is uncertain.

- Probability is a rigorous formalism for uncertain knowledge
- Joint probability distribution specifies the probability of every atomic event
- Queries can be answered by summing over atomic events
- For nontrivial domains, we must find a way to reduce the joint size
- Independence and conditional independence provide the tools

BAYES' RULE AND ITS USE

Bayes' Theorem

Consider the conditional probability statement

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(B|A) = \frac{P(B \cap A)}{P(A)}$$

Bayes' Theorem

The image shows a handwritten derivation of Bayes' Theorem on a whiteboard. The text is written in red ink. At the top left, the formula $P(A|B) = \frac{P(A \cap B)}{P(B)}$ is written, with a red arrow pointing to the numerator. To the right of this, a large curly bracket groups the following equations: $P(A \cap B) = P(B \cap A)$, $P(A \cap B) = P(A|B) * P(B)$, and $P(B \cap A) = P(B|A) * P(A)$. A double-headed red arrow connects the two expressions for $P(A \cap B)$. Below the first equation, the formula $P(B|A) = \frac{P(B \cap A)}{P(A)}$ is written, with a red arrow pointing to the numerator. To the right of this, another large curly bracket groups the equations $P(A|B) * P(B) = P(B|A) * P(A)$ and the final boxed formula $P(A|B) = \frac{P(B|A) * P(A)}{P(B)}$. The final formula is enclosed in a red rectangular box.

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$
$$P(A \cap B) = P(B \cap A)$$
$$P(A \cap B) = P(A|B) * P(B)$$
$$P(B \cap A) = P(B|A) * P(A)$$
$$P(B|A) = \frac{P(B \cap A)}{P(A)}$$
$$P(A|B) * P(B) = P(B|A) * P(A)$$
$$P(A|B) = \frac{P(B|A) * P(A)}{P(B)}$$

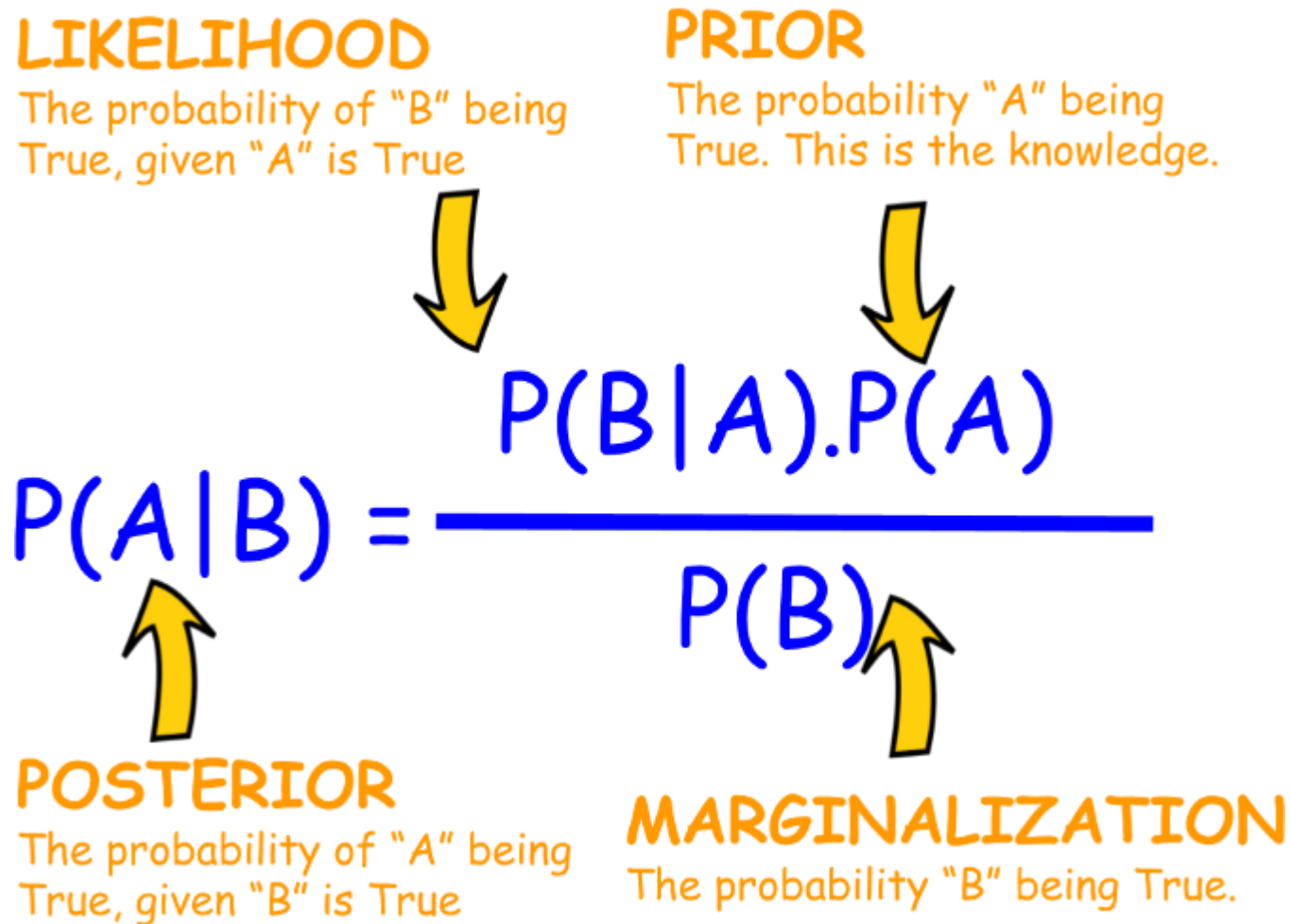
Bayes' Theorem

LIKELIHOOD

The probability of "B" being True, given "A" is True

PRIOR

The probability "A" being True. This is the knowledge.



The diagram illustrates the components of Bayes' Theorem and their relationship to the formula. The formula is $P(A|B) = \frac{P(B|A).P(A)}{P(B)}$. Four yellow arrows point from the component labels to the formula: one from 'LIKELIHOOD' to $P(B|A)$, one from 'PRIOR' to $P(A)$, one from 'POSTERIOR' to $P(A|B)$, and one from 'MARGINALIZATION' to $P(B)$.

$$P(A|B) = \frac{P(B|A).P(A)}{P(B)}$$

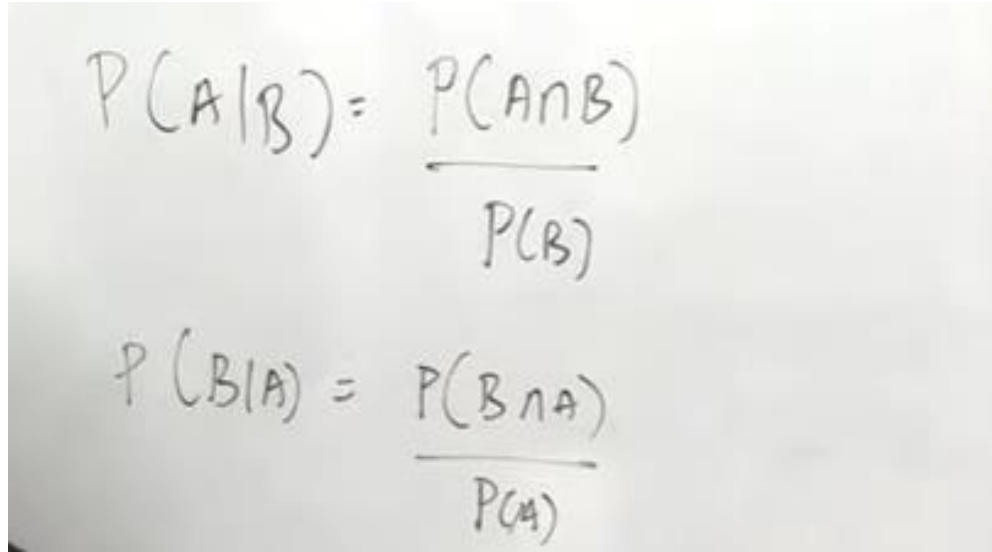
POSTERIOR

The probability of "A" being True, given "B" is True

MARGINALIZATION

The probability "B" being True.

Independent Events (IE)



The image shows two handwritten formulas on a piece of paper. The first formula is $P(A|B) = \frac{P(A \cap B)}{P(B)}$. The second formula is $P(B|A) = \frac{P(B \cap A)}{P(A)}$. Both formulas are written in a cursive, handwritten style.

- Consider A, B - IE
- $P(A \cap B) = P(A) \cdot P(B)$ – apply in conditional probability
- $P(A/B) = P(A) \cdot P(B) / P(B)$ (note: $P(A) \cdot P(B)$ – independent)
- $P(A/B) = P(A)$

Bayes - Example

- A factory has two machines I and II.
- Machine I and II produce 30% and 70% of items respectively.
- Further 3% of items produced by Machines I are defective and 4% of items produced by Machines II are defective.
- An item is drawn at random. If the drawn item is defective, find the probability that it was produced by Machine II.

Solution

Events Definition

- E_1 : Event that the item is produced by Machine I.
- E_2 : Event that the item is produced by Machine II.
- D : Event that the chosen item is defective.

Given Probabilities

- $P(E_1) = 30\% = 0.30$
- $P(E_2) = 70\% = 0.70$
- $P(D \mid E_1) = 3\% = 0.03$
- $P(D \mid E_2) = 4\% = 0.04$

Step-by-Step Calculation

We need to find $P(E_2 \mid D)$, which is the probability that the defective item was produced by Machine II.

Using Bayes' Theorem:

$$P(E_2 \mid D) = \frac{P(E_2) \cdot P(D \mid E_1)}{P(E_1) \cdot P(D \mid E_1) + P(E_2) \cdot P(D \mid E_2)}$$

Substitute the known values into the equation:

$$P(E_2 | D) = \frac{0.70 \cdot 0.04}{(0.30 \cdot 0.03) + (0.70 \cdot 0.04)}$$

$$P(E_2 | D) = \frac{0.028}{0.009 + 0.028}$$

$$P(E_2 | D) = \frac{0.028}{0.037} = \frac{28}{37}$$

Final Result

The probability that the defective item was produced by Machine II is $\frac{28}{37}$ (or approximately 0.7568 / 75.68%).

Simple Example Problem

- A doctor knows that the disease meningitis causes the patient to have a stiff neck, say, 50% of the time. The doctor also knows some unconditional facts: the prior probability that a patient has meningitis is 1/50,000, and the prior probability that any patient has a stiff neck is 1/20. Let **S** be the proposition that the patient has a stiff neck, and **M** be the proposition that the patient has meningitis.

$$P(S|M) = 0.5$$

$$P(M) = 1/50000$$

$$P(S) = 1/20$$

$$P(M|S) = \frac{P(S|M)P(M)}{P(S)} = \frac{0.5 \times 1/50000}{1/20} = 0.0002$$

Naïve Bayes Algorithm

- Naïve Bayes is a **probabilistic machine learning algorithm based on the Bayes Theorem**, used in a wide variety of **classification tasks**.
- **Bayes Theorem**
 - Bayes' Theorem is a simple mathematical formula used for calculating **conditional probabilities**.
- **Conditional probability** is a measure of the probability of an event occurring given that another event has (by assumption, presumption, assertion, or evidence) occurred.

Bayes Theorem Formula

Diagram illustrating the Bayes Theorem formula with labels for each component:

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Labels and their corresponding parts of the formula:

- Probability of A occurring given evidence B has already occurred** points to $P(A|B)$.
- Probability of B occurring given evidence A has already occurred** points to $P(B|A)$.
- Probability of A occurring** points to $P(A)$.
- Probability of B occurring** points to $P(B)$.

Naïve Bayes - EXAMPLE PROBLEM

Class-labeled training tuples from the *AllElectronics* customer database.

<i>RID</i>	<i>age</i>	<i>income</i>	<i>student</i>	<i>credit_rating</i>	<i>Class: buys_computer</i>
1	youth	high	no	fair	no
2	youth	high	no	excellent	no
3	middle_aged	high	no	fair	yes
4	senior	medium	no	fair	yes
5	senior	low	yes	fair	yes
6	senior	low	yes	excellent	no
7	middle_aged	low	yes	excellent	yes
8	youth	medium	no	fair	no
9	youth	low	yes	fair	yes
10	senior	medium	yes	fair	yes
11	youth	medium	yes	excellent	yes
12	middle_aged	medium	no	excellent	yes
13	middle_aged	high	yes	fair	yes
14	senior	medium	no	excellent	no

The training data are in above Table. The data tuples are described by the attributes age, income, student, and credit rating. The class label attribute, buys computer, has two distinct values (namely, yes, no). Let C_1 correspond to the class buys computer = **yes** and C_2 correspond to buys computer = **no**. The tuple we wish to classify is **$X = (\text{age} = \text{youth}, \text{income} = \text{medium}, \text{student} = \text{yes}, \text{credit rating} = \text{fair})$**

- We need to maximize $P(X|C_i)P(C_i)$, for $i = 1, 2$. $P(C_i)$, the prior probability of each class, can be computed based on the training tuples:
 - **$P(\text{buys computer} = \text{yes}) = 9/14 = 0.643$**
 - **$P(\text{buys computer} = \text{no}) = 5/14 = 0.357$**
- To compute $P(X|C_i)$, for $i = 1, 2$, we compute the following conditional probabilities:

- $P(\text{age} = \text{youth} \mid \text{buys computer} = \text{yes}) = ?$
- $P(\text{age} = \text{youth} \mid \text{buys computer} = \text{no}) = ?$
- $P(\text{income} = \text{medium} \mid \text{buys computer} = \text{yes}) = ?$
- $P(\text{income} = \text{medium} \mid \text{buys computer} = \text{no}) = ?$
- $P(\text{student} = \text{yes} \mid \text{buys computer} = \text{yes}) = ?$
- $P(\text{student} = \text{yes} \mid \text{buys computer} = \text{no}) = ?$
- $P(\text{credit rating} = \text{fair} \mid \text{buys computer} = \text{yes}) = ?$
- $P(\text{credit rating} = \text{fair} \mid \text{buys computer} = \text{no}) = ?$
- **The above full join distribution can be written as**

$$P(\text{Cause}, \text{Effect}_1, \dots, \text{Effect}_n) = P(\text{Cause}) \prod_i P(\text{Effect}_i \mid \text{Cause})$$

$$P(\text{age} = \text{youth} \mid \text{buys_computer} = \text{yes}) = 2/9 = 0.222$$

$$P(\text{age} = \text{youth} \mid \text{buys_computer} = \text{no}) = 3/5 = 0.600$$

$$P(\text{income} = \text{medium} \mid \text{buys_computer} = \text{yes}) = 4/9 = 0.444$$

$$P(\text{income} = \text{medium} \mid \text{buys_computer} = \text{no}) = 2/5 = 0.400$$

$$P(\text{student} = \text{yes} \mid \text{buys_computer} = \text{yes}) = 6/9 = 0.667$$

$$P(\text{student} = \text{yes} \mid \text{buys_computer} = \text{no}) = 1/5 = 0.200$$

$$P(\text{credit_rating} = \text{fair} \mid \text{buys_computer} = \text{yes}) = 6/9 = 0.667$$

$$P(\text{credit_rating} = \text{fair} \mid \text{buys_computer} = \text{no}) = 2/5 = 0.400$$

$$P(\text{Cause}, \text{Effect}_1, \dots, \text{Effect}_n) = P(\text{Cause}) \prod_i P(\text{Effect}_i \mid \text{Cause})$$

$$\begin{aligned} P(X \mid \text{buys_computer} = \text{yes}) &= P(\text{age} = \text{youth} \mid \text{buys_computer} = \text{yes}) \times \\ &\quad P(\text{income} = \text{medium} \mid \text{buys_computer} = \text{yes}) \times \\ &\quad P(\text{student} = \text{yes} \mid \text{buys_computer} = \text{yes}) \times \\ &\quad P(\text{credit_rating} = \text{fair} \mid \text{buys_computer} = \text{yes}) \end{aligned}$$

$$= 0.222 \times 0.444 \times 0.667 \times 0.667 = \mathbf{0.044}$$

Similarly,

$$P(X \mid \text{buys computer} = \text{no})$$

$$= 0.600 \times 0.400 \times 0.200 \times 0.400 = \mathbf{0.019}$$

To find the class, C_i , that maximizes $P(X \mid C_i)P(C_i)$, we compute

$$P(X \mid \text{buys computer} = \text{yes})P(\text{buys computer} = \text{yes}) = ?$$

$$P(X \mid \text{buys computer} = \text{no})P(\text{buys computer} = \text{no}) = ?$$

$$P(X \mid \text{buys computer} = \text{yes})P(\text{buys computer} = \text{yes}) \\ = 0.044 \times 0.643 = \mathbf{0.028}$$

$$P(X \mid \text{buys computer} = \text{no})P(\text{buys computer} = \text{no}) \\ = 0.019 \times 0.357 = \mathbf{0.007}$$

Therefore, the naïve Bayesian classifier predicts **buys computer = yes** for tuple **X**.

Outlook	Temperature	Humidity	Windy	Play
sunny	hot	high	false	no
sunny	hot	high	true	no
overcast	hot	high	false	yes
rainy	mild	high	false	yes
rainy	cool	normal	false	yes
rainy	cool	normal	true	no
overcast	cool	normal	true	yes
sunny	mild	high	false	no
sunny	cool	normal	false	yes
rainy	mild	normal	false	yes
sunny	mild	normal	true	yes
overcast	mild	high	true	yes
overcast	hot	normal	false	yes
rainy	mild	high	true	no